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eGramSabha

AI for Bharat Hackathon

AI-Powered Digital Gram Sabha Platform

Powered by aws

Team Name: Empower Panchayat

Team Leader Name: (see submission form)

Problem Statement: AI-Powered Digital Gram Sabha Platform

Problem Statement

India has 250,000+ Gram Panchayats. The 73rd Constitutional Amendment (1992) mandates Gram Sabhas as the foundation of grassroots democracy. Three decades later, the promise remains unfulfilled.

5-10%
Average attendance at Gram Sabhas

<3%
Women participation nationwide

60%+
Citizens with limited literacy

30%
Excluded from welfare schemes

ZERO
Digital trail of decisions

Barrier	Impact
Paper agendas on notice boards	Low awareness, excludes remote hamlets
Paper sign-in sheets	Easy to forge, no biometric verification
Hand-written minutes	Subjective, incomplete, manipulable
Single language proceedings	Excludes tribal/minority communities
No issue tracking	Promises forgotten between meetings
Physical presence only	Excludes women, elderly, differently-abled

Brief About the Idea

Solution Overview

eGramSabha is a voice-first, face-authenticated platform that digitizes the entire Gram Sabha lifecycle.

Why AI is Required

60%+ rural Indians have limited literacy -- voice-first AI removes the single largest barrier to democratic participation

AI transcribes, corrects, translates citizen voice complaints into formal issues in 3 languages

AI auto-detects urgency -- health/safety issues flagged URGENT automatically via sentiment analysis

AI generates Minutes of Meeting -- eliminates subjective hand-written minutes, creates permanent digital record

AI generates structured agendas from citizen issues -- democratic by default, not influenced by powerful voices

Text-to-Speech reads agendas aloud -- accessibility for citizens who cannot read

Without AI, this platform would require an army of clerks, translators, and transcribers at every panchayat

Brief About the Idea (continued)

AWS Services (11 Active + 4 Code-Ready = 15)

Bedrock (Nova Lite / Claude Sonnet): LLM backbone: MOM, Agenda, Grammar, Sentiment, Translation fallback
S3 (2 buckets, Intelligent-Tiering): Face images, TTS audio cache, attachments
Translate: Real-time translation across 10+ Indian languages
Polly: TTS for agenda read-aloud (Aditi Hindi, Raveena English, S3 cached)
API Gateway: Single entry point, 100 req/s rate limiting, API key management
CloudWatch: Centralized structured logging (Node.js + Python)
Secrets Manager: Zero hardcoded credentials, JWT secrets at runtime
EC2 + EBS + Elastic IP: Compute, storage, static networking
IAM: Least-privilege policies per service

Code-Ready:
Rekognition: Face verification (90% threshold)
DocumentDB: Managed MongoDB migration
Transcribe: AWS STT (28 languages)
SNS: SMS notifications (Transactional)

User	Without AI	With AI
Illiterate citizen	Cannot report issues (requires writing)	Speaks complaint, AI handles the rest
Panchayat secretary	Hand-writes minutes for hours	AI generates MOM in seconds, in 3 languages
Women at home	Must attend male-dominated public meeting	Reports issues via voice from phone
Official creating agenda	Ad-hoc, influenced by powerful voices	AI structures agenda from all citizen issues
Tribal citizen	Excluded by state language proceedings	Speaks native language, AI translates
Block/District admin	Physical inspection needed	Digital records, AI analytics, remote audit

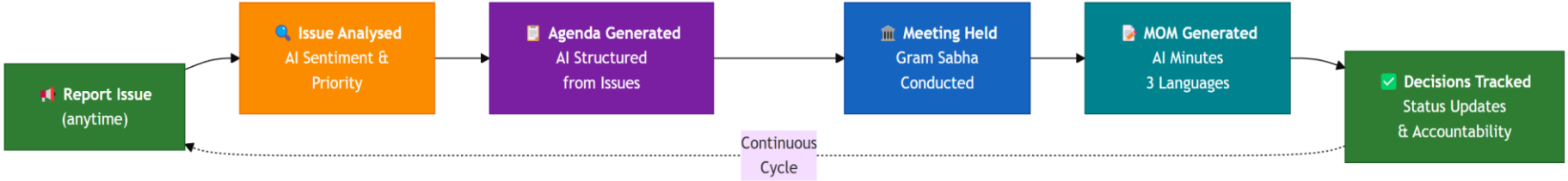
List of Features

- Facial Recognition Citizen Authentication (liveness detection: blink + head movement)
- Voice-First Issue Reporting (record complaint in any language)
- AI Transcription + Grammar Correction + Translation (3 languages)
- AI Sentiment Analysis + Key Phrase Extraction + Auto-Priority
- AI-Powered Agenda Generation from citizen issues
- AI-Generated Minutes of Meeting (structured, multilingual)
- Text-to-Speech Agenda Read-Aloud (Amazon Polly, S3 cached)
- SMS Notifications (Amazon SNS, code-ready)
- Biometric Attendance Tracking (face + manual)
- Meeting RSVP Management
- Multi-Portal System (Admin, Official, Citizen)
- Role-Based Access Control (6 roles, 5 resources)
- Bulk CSV Citizen Import
- Custom Panchayat Letterhead for PDFs
- Multi-Language UI (English + Hindi)
- Support Ticket System (anonymous allowed)
- PDF Export (attendance, MOM, reports)
- Provider Abstraction (swap any service via env var)

Visual Representations

5-Step Gram Sabha Lifecycle

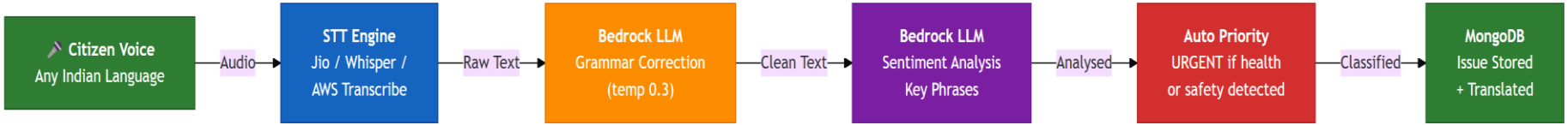
Continuous Governance Loop



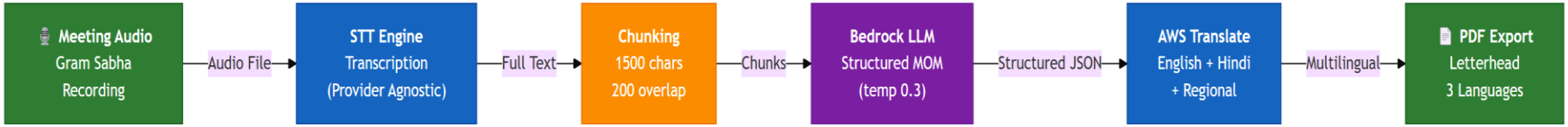
Citizens can report issues at any time -- the cycle is continuous, not tied to meeting dates

Process Flow: AI Pipelines

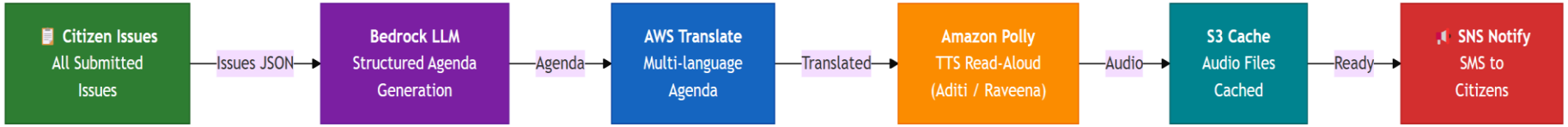
Voice Issue Reporting *(cronJobs.js:51-53)*



MOM Generation *(llm_service.py:763-791)*

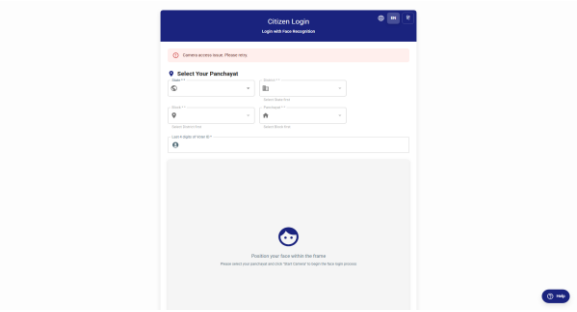


Agenda Generation *(llm_service.py)*

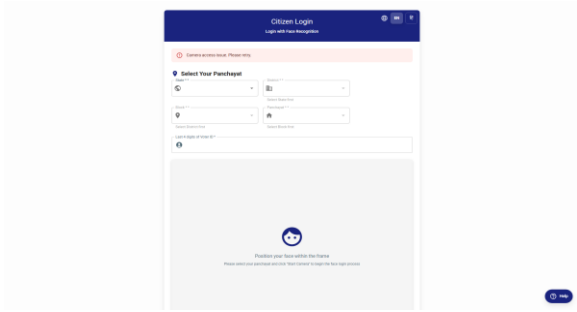


Face Authentication *(citizenAuthRoutes.js:210)*

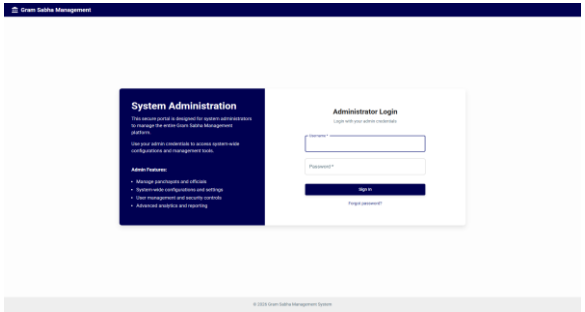
Wireframes / Prototype Screenshots



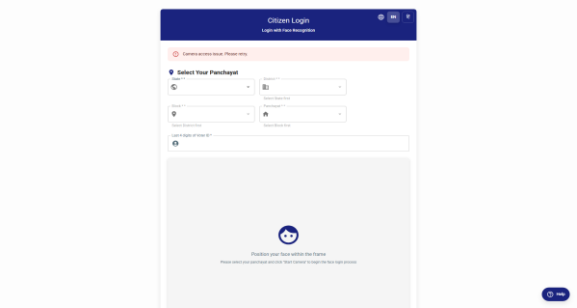
Citizen Login Flow (Voter ID + Face Scan + Liveness)



Citizen Dashboard (Issue Stats, Upcoming Meetings)

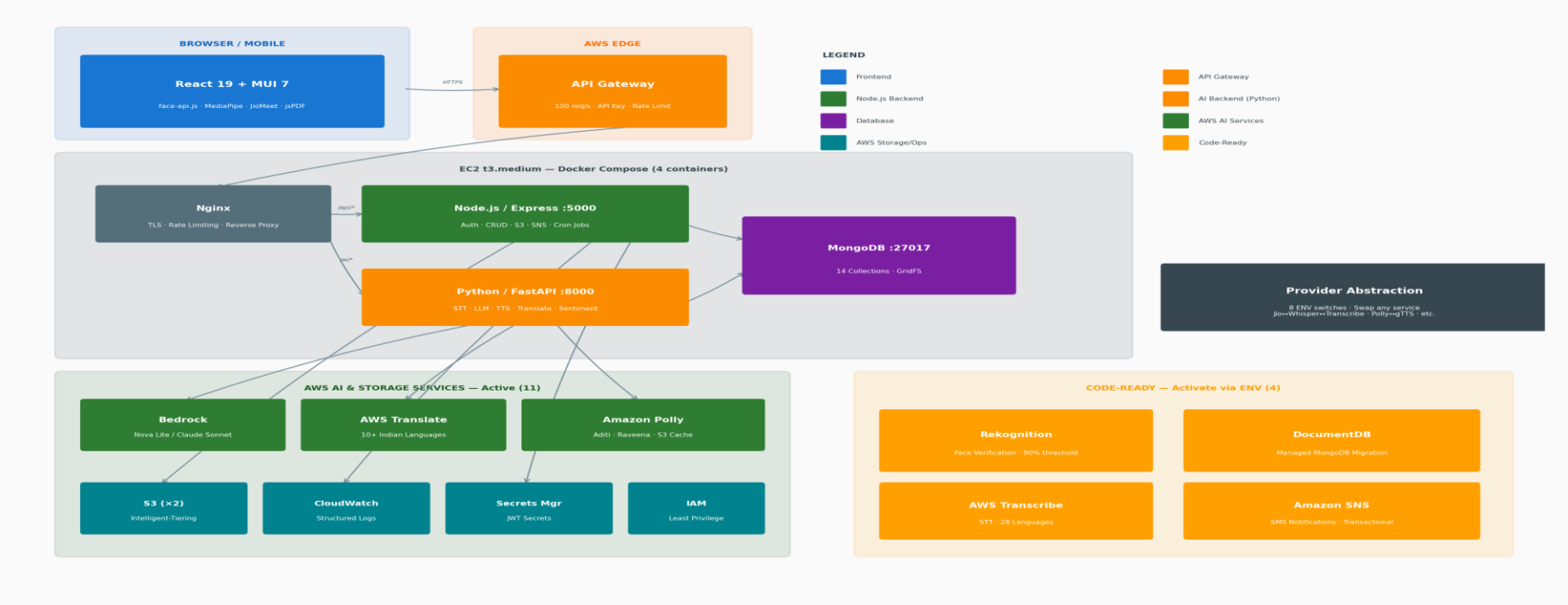


Voice Issue Reporting (Audio Recorder, Attachments)



Issue List (Sentiment Badges, Priority, Transcription)

Architecture Diagram



Source: docker-compose.prod.yml, infra/setup-api-gateway.js, frontend/nginx.prod.conf | 4 Docker containers on bridge network | 11 Active + 4 Code-Ready AWS services

Technologies Utilized

Layer	Technology	Purpose
Frontend	React 19, MUI 7, face-api.js, MediaPipe	SPA with face recognition
Backend API	Node.js, Express, Mongoose 8	CRUD, auth, file management
AI Backend	Python, FastAPI, Motor (async)	AI pipelines, async processing
Database	MongoDB 7 (DocumentDB-ready)	14 collections, GridFS
AI/LLM	Amazon Bedrock (Converse API)	MOM, agenda, grammar, sentiment
STT	Jio STT / Whisper / AWS Transcribe	Speech-to-text (provider-agnostic)
Translation	AWS Translate	10+ Indian languages
TTS	Amazon Polly + S3 cache	Agenda read-aloud
Storage	Amazon S3 (Intelligent-Tiering)	Face images, audio, attachments
Security	Secrets Manager, JWT, Helmet, CORS	Zero hardcoded credentials
Traffic	API Gateway, Nginx reverse proxy	Rate limiting, TLS, API keys
Logging	CloudWatch (winston + watchtower)	Centralized structured logs
Infra	Docker Compose, EC2, EBS, Elastic IP	Container orchestration
Docs	jsPDF, html2pdf.js	PDF generation with Hindi fonts
Video	JioMeet SDK	Virtual meeting integration

Estimated Implementation Cost

Current Hackathon Cost: $15-20/month

Service	Monthly Cost	Notes
EC2 (t3.medium)	$12-30	On-demand vs reserved
EBS (20 GB gp3)	$1.60	
S3	$0.50	Minimal storage
Bedrock	$2-5	Pay per token
Translate	$0-1	2M chars/mo free tier
Polly	$0-0.50	5M chars/mo free tier
CloudWatch	$0-0.50	5 GB/mo free tier
Secrets Manager	$0.40	1 secret
API Gateway	$0	1M calls/mo free tier
TOTAL	$15-20/month	

Scale Projections

Phase	Scale	Monthly	Per Citizen/Yr
Hackathon	1-5 panchayats	$15-20	--
Pilot	100 panchayats	$320	~$3
State	5K panchayats	$3,250	~$0.80
National	250K panchayats	$30-37K	$0.60-0.72

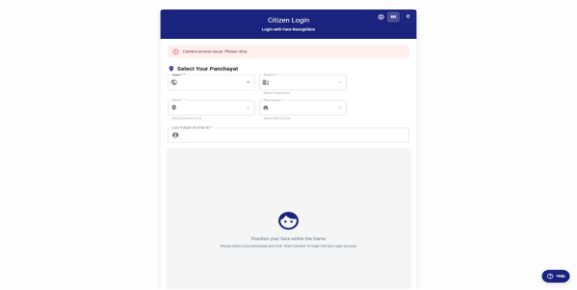
~100x Cost Reduction

Rs 500+/meeting (physical) vs Rs 5/citizen/year (digital)

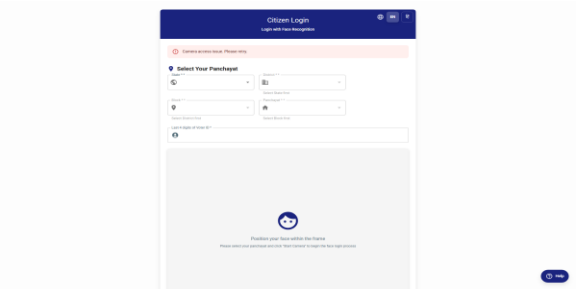
Cost Optimizations:

S3 Intelligent-Tiering for auto storage class
Polly S3 caching (generate once, serve many)
API Gateway throttling (100 req/s)
Bedrock low temperature (0.3) for efficiency

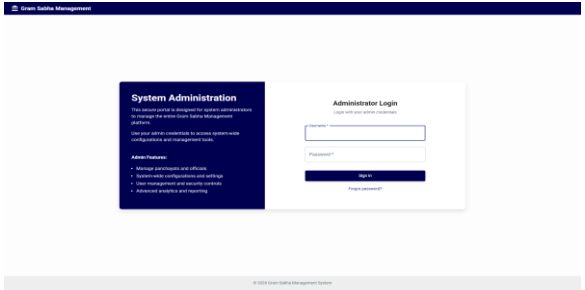
Snapshots of the Prototype



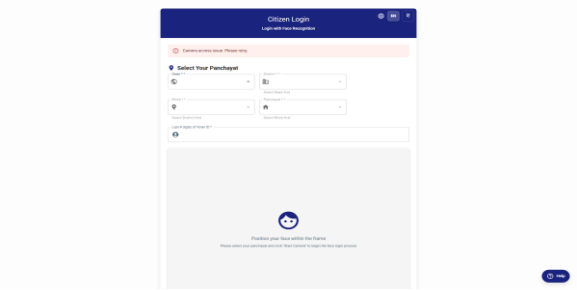
Transcription: Original -> English -> Hindi output



Sentiment Analysis badges with confidence scores



MOM output with structured sections



Attendance tracking with face verification

Read Aloud button and TTS playback

PDF export with panchayat letterhead

Prototype Performance Report

Tatarpur Digital Gram Sabha (April 2025) -- India's First

Metric	National Avg	Tatarpur Digital	Improvement
Voter registration	~10% active	81% (650/801)	8x
Issues submitted	Handful (verbal)	100+ in 3 days	10x+
Women participation	~3%	~50%	16x
Time to minutes	Days (if ever)	Minutes (AI)	Instant

Key outcomes: garbage management system, road repairs, transgender welfare (Rs 2,100 min), solar plant on 24 acres

Impact Summary

Impact Area	Mechanism	Outcome
Citizen Participation	Voice-first, face-auth	5% -> 30%+ attendance
Women's Empowerment	Remote voice reporting	3% -> 50% participation
Welfare Delivery	Citizen-verified lists	Reduce 30% exclusion
Transparency	AI-generated MOMs	Every decision verifiable
Accountability	Issue lifecycle tracking	Zero promises lost
Cost Reduction	Digital vs paper	~100x reduction

Source: IMPACT.md:14,330-340 | Tatarpur Digital Gram Sabha (April 2025)

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Additional Details / Future Development

Immediate

- Activate AWS Transcribe for managed STT
- DLT registration for SNS SMS notifications
- Enable Amazon Rekognition for face verification

Short-term

- DocumentDB migration for managed MongoDB
- Native mobile app (React Native)
- e-Gram Swaraj + Panchayat integration

Medium-term

- Multi-panchayat escalation workflows
- Block/District analytics dashboards
- Gram Sabha with AI [illegible]

Long-term

- National rollout (250K panchayats)
- Aadhaar integration for identity
- Open API for third-party integrations
- Scheme delivery tracking & verification

Powered by aws

Thank You

"Sashakt Panchayat, Samriddh Bharat" -- Strong Panchayat, Prosperous India

Digital India alignment: UPI, Aadhaar, DBT, CoWIN pattern

GitHub Repository: https://github.com/anshuljain90/ai4bharat-empowerpanchayat

Live Deployment: https://empowerpanchayat.org/

Demo Video: (Max 3 minutes -- link to be provided)

License: Apache 2.0 -- Digital Public Good (DPG)